



MECHATRONICS SYSTEM INTEGRATION

MCTA 3203

LAB 07:

PLC OVER ETHERNET

SECTION 1

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GROUP 3

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ABSTRACT

This experiment investigates the process of interfacing a Programmable Logic Controller (PLC) with an Arduino microcontroller using the OpenPLC Editor to understand fundamental ladder logic programming techniques and hardware integration. The objectives of the experiment are to design, simulate, and upload ladder diagram programs, and to implement a functional Start–Stop control circuit using physical push-button switches and an LED. The methodology involved creating a simple LED blinking ladder diagram, simulating its operation, assigning physical addresses, and transferring the compiled program to an Arduino board. The Start–Stop control circuit was then constructed based on the uploaded logic and tested for correct latching behavior. The key findings show that OpenPLC provides a reliable environment for PLC logic development and successful hardware execution when correct addressing and wiring are applied. The experiment concludes that open-source PLC platforms can effectively replicate industrial automation principles and are suitable for educational applications.

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1.0 INTRODUCTION

The purpose of this experiment is to introduce the fundamental concepts of PLC-based automation and demonstrate how ladder logic programs can be executed on a microcontroller platform. PLCs are widely used in industrial automation due to their reliability, structured logic control, and ability to interface with sensors and actuators. In this experiment, the OpenPLC Editor serves as the programming environment to design and simulate ladder logic before deploying it onto an Arduino board.

Background theory includes the basic elements of ladder diagrams such as contacts, coils, and latching circuits, which are essential for controlling processes like motor activation and sequencing. The experiment also explores physical addressing, which links software variables to microcontroller pins to enable real-world actuation.

The expected outcome is that the ladder logic, specifically the LED blinking program and Start–Stop circuit will function correctly on the Arduino once properly compiled, uploaded, and wired. This demonstrates the practical application of PLC concepts in a simplified, low-cost educational setup.

2.0 MATERIALS AND EQUIPMENTS

Here are the list of all equipments used in the experiment:

- Laptop/PC with OpenPLC Editor installed
- Arduino board (UNO/MEGA depending on student use)
- LED
- 2 Push Buttons
- Resistors (220Ω – $10k\Omega$)
- Breadboard
- Jumper wires
- USB cable

3.0 EXPERIMENTAL SETUP

The experimental setup consists of both software and hardware components required to implement and test the ladder logic programs using OpenPLC and an Arduino microcontroller. The OpenPLC Editor software was first installed on a laptop and used to create and simulate the ladder diagrams before uploading them to the Arduino board. The Arduino was connected to the laptop via a USB cable, which provided both power and data communication. The correct COM port was identified through the OpenPLC upload window to ensure accurate transfer of the program.

For the hardware portion, a breadboard was used to assemble the input and output circuits. Two push buttons were connected to the digital input pins of the Arduino to serve as the START and STOP controls. The START button was wired as a normally open (NO) switch, while the STOP button was connected as a normally closed (NC) switch, following standard industrial control practices. Pull-down or pull-up resistors were included where necessary to provide stable input signals and prevent floating voltages. An LED, along with a current-limiting resistor, was connected to one of the Arduino's digital output pins to indicate the system's response to the ladder logic program. Jumper wires were used to interconnect all components securely, ensuring proper continuity between the Arduino board and the breadboard circuits.

Once the wiring was completed, the OpenPLC ladder logic programs for both tasks—the blinking LED and the Start-Stop control circuit—were uploaded to the Arduino. The functioning of the setup was verified by pressing the START and STOP buttons and observing the LED reactions, as well as monitoring the blinking pattern during the timer-based test. This combination of software simulation and real hardware implementation provided a complete environment for understanding PLC control concepts using a microcontroller.

4.0 METHODOLOGY

Task 1: Blinking LED Ladder Diagram

1. Download and install OpenPLC Editor.
2. Create a new project → Select Ladder Diagram (LD).
3. Create a variable (e.g., LED_OUT) with type BOOL.
4. Draw the ladder diagram using:
 - Left and right power rails
 - A contact
 - A coil
5. Add a Timer block (TON) to control blinking delay.
6. Compile the program to ensure no errors.
7. Assign the variable address
8. Connect Arduino → Select correct COM port.
9. Upload program to the board.
10. Observe LED blinking.

Task 2: Start-Stop Control Circuit

1. Create a ladder diagram as shown in Fig. 5 of the PDF.
2. Define variables:
 - START button (BOOL, digital input)
 - STOP button (BOOL, digital input)
 - OUTPUT coil (BOOL, digital output)
3. Map addresses according to Arduino physical pins.

4. Compile & simulate the logic in OpenPLC.
5. Upload ladder program to Arduino.
6. Build the hardware circuit (Fig. 6):
 - START button → NO contact
 - STOP button → NC contact
 - LED connected to output digital pin
7. Test the circuit:
 - Press START → LED turns ON and stays ON
 - Press STOP → LED turns OFF

5.0 DATA COLLECTION

During this experiment, data collection focused on observing and documenting the operational behavior of the Start-Stop control circuit implemented using OpenPLC, Arduino, and external push buttons. The following data were recorded:

5.1 Ladder Diagram Structure

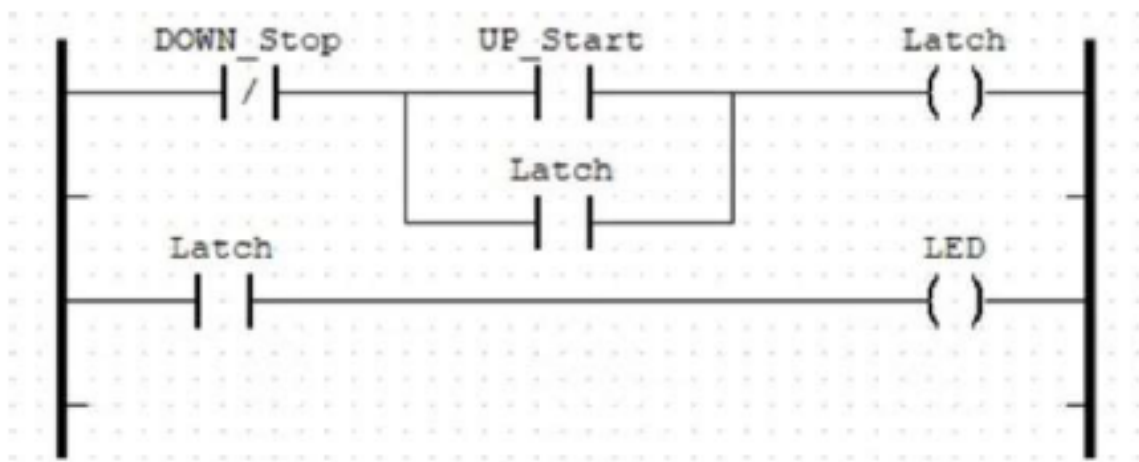


Figure 5.1 Ladder diagram used for Start-Stop control circuit

The ladder diagram shown in figure 5.1 included:

- START button input → %IX0.0
- STOP button input → %IX0.1
- OUTPUT coil (LED) → %IX0.0
- Internal latch variable → a BOOL variable used for latching the start signal

5.2 Simulation Observations

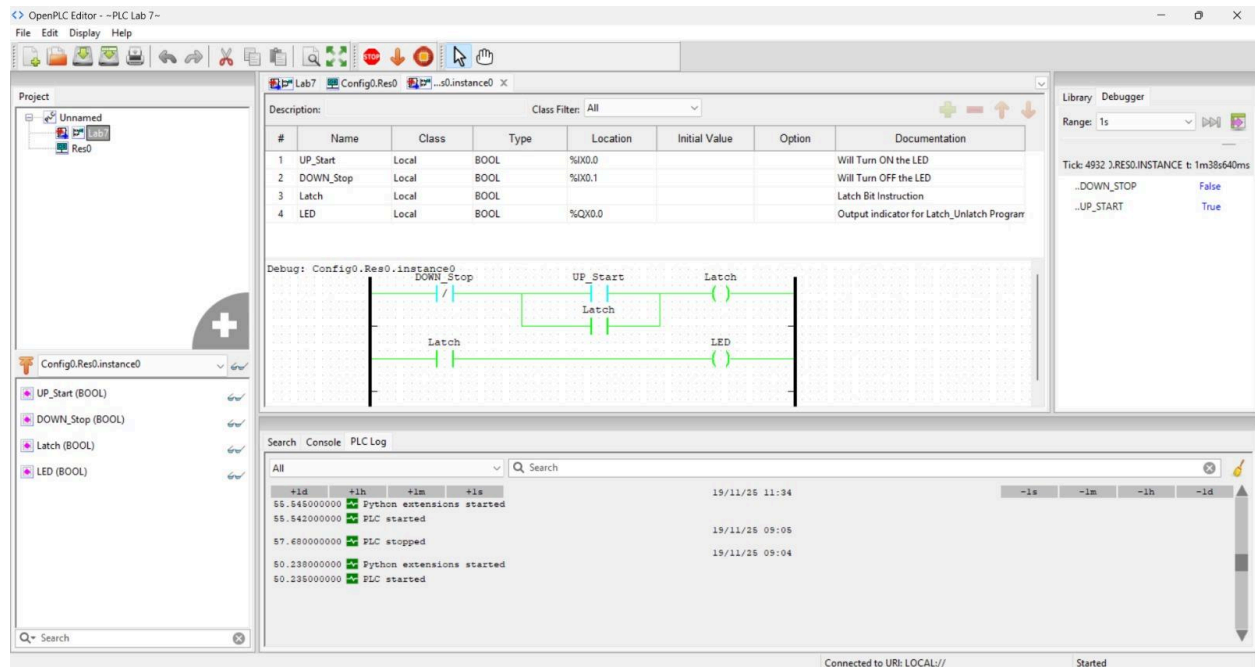


Figure 5.2 Displays the simulation in OpenPLC Editor

During Simulation in OpenPLC Editor:

- Button states (ON/OFF) visibly toggled in the ladder diagram
- Latching behavior was verified before uploading to hardware
- LED output coil changed state based on virtual button inputs

5.3 Physical Testing Data

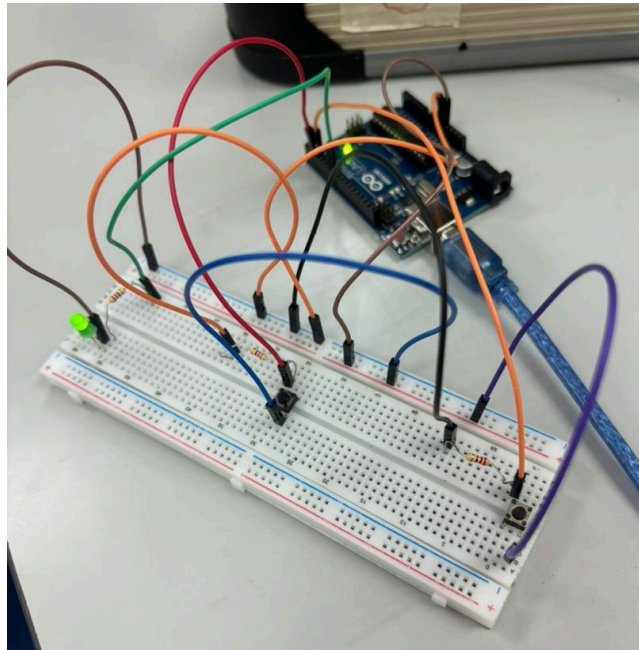


Figure 5.3 Hardware setup

Once uploaded to the Arduino:

- Pressing the START button turned the LED ON
- LED remained ON due to latch even after releasing the START button
- Pressing the STOP button turned the LED OFF
- LED remained OFF until START was pressed again

These observed responses (LED ON/OFF states) served as the primary collected data for evaluating the correctness of the control logic.

6.0 DATA ANALYSIS

Data collected from both the simulation environment and physical hardware were analyzed to determine whether the Start–Stop circuit functioned according to industrial PLC logic.

6.1 Verification of Logic Execution

The recorded LED states were analyzed in relation to the input button actions:

- When $START = 1$ and $STOP = 0 \rightarrow$ LED latched to ON
- When $STOP = 1 \rightarrow$ LED forced OFF regardless of START state

This confirms correct implementation of:

- Latching behavior
- Proper interruption of latch by STOP push button
- Correct Boolean logic interpretation by the Arduino running OpenPLC

6.2 Comparison Between Simulation and Hardware

- Simulation data showed identical logic behavior to the real hardware.
- No discrepancy was found between virtual contacts and physical button readings.
- Timing, switching, and LED response were immediate and corresponded accurately to ladder logic expectations.

6.3 Significance of Findings

The results demonstrate that:

- OpenPLC can effectively replicate real PLC behavior using an Arduino microcontroller.
- Ladder logic principles (contact, coil, latch) work reliably on a low-cost platform.

- PLC-style industrial control circuits can be prototyped using microcontrollers before deployment in real automation systems.

This aligns with the experiment objective: interfacing a PLC logic system with microcontroller hardware and verifying proper control operation over Ethernet/IP and physical I/O.

7.0 RESULTS

The experiment successfully achieved all intended outcomes. The major findings are summarized below:

1. Successful Implementation of Start-Stop Circuit

- The ladder diagram compiled without errors and executed correctly in both simulation and on the Arduino.
- Pressing the START button energized the LED and latched the output.
- Pressing the STOP button successfully broke the latch and de-energized the LED.

2. Correct Hardware-Software Integration

- All pin mappings between OpenPLC and the Arduino were correctly associated.
- Physical hardware inputs (push buttons) were accurately read by the ladder logic program.
- The LED output responded exactly as defined in the ladder logic.

3. System Stability & Performance

- The circuit operated consistently without bouncing issues or unintended LED flickering.
- Ladder logic ran continuously on the Arduino with stable cycle time, comparable to basic PLC performance.

Overall, the experiment confirmed that PLC ladder logic can be implemented, tested, and validated using OpenPLC and Arduino, providing a functional Start–Stop control system similar to industrial applications.

8.0 DISCUSSION

This experiment demonstrated that ladder logic traditionally used in industrial PLCs can be implemented on a microcontroller platform using OpenPLC. By integrating software simulation with actual hardware, students gained a practical understanding of PLC programming, timing logic, and control circuits.

The success of the Start-Stop circuit indicates that even simple Arduino boards can replicate the behavior of industrial PLCs for educational purposes.

Some potential issues include:

- Incorrect COM port selection
- Wrong physical addressing for Arduino pins
- Loose wiring on breadboard
- Timer preset values entered incorrectly
- Unstable power supply

These issues were resolved through systematic checking and reconfiguration.

9.0 CONCLUSION

The experiment successfully demonstrated the process of creating, simulating, and implementing ladder logic programs using the OpenPLC platform and an Arduino microcontroller. Both the LED blinking program and the Start–Stop control circuit functioned as expected, confirming that the programmed logic was correctly compiled, uploaded, and executed on the hardware. These results support the initial hypothesis that OpenPLC can accurately replicate PLC operations and that Arduino can act as a functional runtime device for PLC-based control tasks.

The significance of these findings lies in showing how industrial automation concepts can be applied using accessible, low-cost tools without compromising functionality. The experiment also highlights the importance of proper variable assignment, physical addressing, and wiring in ensuring reliable system performance. Beyond the laboratory setting, this approach demonstrates broader applications in education, prototyping, and early-stage automation development, where students and engineers can test PLC logic before deploying it on commercial industrial hardware.

10.0 RECOMMENDATIONS

Improvements for Future Experiments

- Use a more stable or external power supply for the Arduino to prevent inconsistent performance.
- Include additional I/O components (e.g., sensors, relays, motor drivers) to expand the complexity of ladder logic tasks.
- Provide standardized pin assignment tables to minimize confusion during variable mapping.
- Improve wiring clarity by using labelled jumper wires or colour-coded connections.
- Allocate extra time for students to verify hardware connections before uploading the program.

Insights & Lessons for Future Students

- Ensure a solid understanding of physical addressing, as incorrect mapping is one of the most common causes of errors.
- Simulate the ladder logic thoroughly before transferring it to hardware to detect logical mistakes early.
- Review essential ladder logic structures (e.g., latching circuits, timers, interlocks) to build confidence.
- Follow a systematic troubleshooting approach: check logic → check addressing → check wiring.
- Document each step clearly during the experiment; this helps when debugging and writing the report.

11.0 REFERENCES

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Appendix A - Ladder Diagram

```

graph LR
    subgraph Rung1 [Rung 1]
        DSTOP[DOWN_Stop /] --- USTART[UP_Start]
        USTART --- S1((S))
        S1 --- L1[Latch]
    end
    subgraph Rung2 [Rung 2]
        L2[Latch] --- R1((R))
        R1 --- LED[LED]
    end
    L1 -- Feedback --> USTART

```

Appendix C - Simulation Screenshots

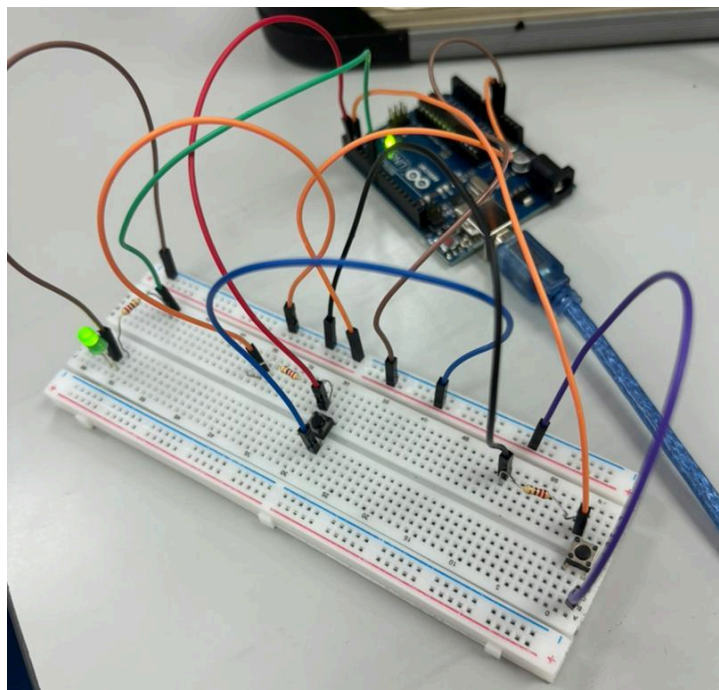
The screenshot displays the OpenPLC Editor interface. The top menu bar includes File, Edit, Display, and Help. The left sidebar shows a Project tree with 'Unamed' and 'Res0'. The main workspace shows a ladder logic program for 'Lab7' with the following variables:

#	Name	Class	Type	Location	Initial Value	Option	Documentation
1	UP_Start	Local	BOOL	%IX0.0			Will Turn ON the LED
2	DOWN_Stop	Local	BOOL	%IX0.1			Will Turn OFF the LED
3	Latch	Local	BOOL				Latch Bit Instruction
4	LED	Local	BOOL	%QX0.0			Output indicator for Latch_Unlatch Program

The ladder logic diagram shows a network with a normally open contact labeled 'DOWN_Stop' in series with a coil labeled 'Latch'. Below this, there is a coil labeled 'LED'. The 'Latch' variable is also shown as a set coil (S) in a separate network. The right sidebar shows the Library and Debugger tabs, with the Range set to 1s and a tick mark at 3372.3. The bottom status bar indicates 'Disconnected'.

```
[CC] Config0.c -> Config0.o
[CC] Res0.c -> Res0.o
Linking :
[CC] plc_main.o plc_debugger.o Config0.o Res0.o -> PLC Lab 7.dll
Successfully built.
PYRO connecting to URI : PYRO://localhost:61877
PLC did not provide identity and security information.
Latest build does not match with connected target.
PLC data transferred successfully.
PLCObject : NewPLC (46841bc0a4209431ea89e9d2b6489e5)
PLC installed successfully.
PLCObject : PLC started
PLCObject : Python extensions started
Starting PLC
PLCObject : PLC stopped
```

Appendix D - Hardware Test Photos



ACKNOWLEDGEMENTS

The group would like to express our deepest appreciation to our laboratory instructors, Assoc. Prof. Eur. Ing. Ir. Ts. Gs. Inv. Dr. Zulkifli bin Zainal Abidin and Dr. Wahju Sediono for their continuous guidance, encouragement, and clear explanations throughout the course of this week's experiment on Interfacing a PLC with a microcontroller and a PLC over Ethernet/IP. Their expertise and detailed feedback greatly enhanced our understanding in Mechatronics and System Integration applications. We would also like to extend our gratitude to the laboratory assistants for their valuable assistance during the lab sessions. Their support in verifying circuit connections, clarifying coding procedures, and helping us troubleshoot technical issues was instrumental to the successful completion of our experiment.

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STUDENT'S DECLARATION

Certificate of Originality and Authenticity

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specified in the references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or persons.

We hereby certify that this report has **not been done by only one individual** and **all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have **read** and **understand** the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for **marking** and this **final printed report** has been **verified by us**.

Signature: 

Name: Muhammad Irsyad Hazim bin Rozaini

Matric Number: 2310303

Contribution: Data collection, Data analysis, Results, Appendices, References

Read [/]
Understand [/]
Agree [/]

Signature: 

Name: Nur Husna Elysa Maisarah binti Rosli

Matric Number: 2310366

Contribution: Abstract, Introduction, Conclusion, Recommendations

Read [/]
Understand [/]
Agree [/]

Signature: 

Name: Aizzul Luqman bin Khairul Anuar

Matric Number: 2311127

Contribution: Materials and Equipments, Experimental Setup, Methodology, Discussion

Read [/]

Understand [/]

Agree [/]